Notebook

March 20, 2021

1 Part 0: Background

Let’s try to understand the background of our dataset before diving into a full-scale analysis.

1.1 Question 0

1.1.1 Part 1

Given that this data compiled by the CCAO (as opposed to other assessor’s offices), find a column that would most likely only be collected in Cook County. What does this column represent?

Neighborhood code is a column that is most likely only to be collected in Cook County, as the description states ”Neighborhood code as assigned by the Assessment office”.

1.1.2 Part 2

Name a feature that isn’t listed in this dataset but may be useful for predicting sales values. What insights could this feature provide? How might it increase or decrease a home’s sales value?

A feature like average salary of the previous accumunated house owners could be beneficial. It could provide insights about how pricey a neighborhood or house is. It might serve as a purpose to stabilize a house price, by trying to match it with it’s priciness of it’s nearby surrounding houses/neighborhood.

1.2 Question 1

1.2.1 Part 1

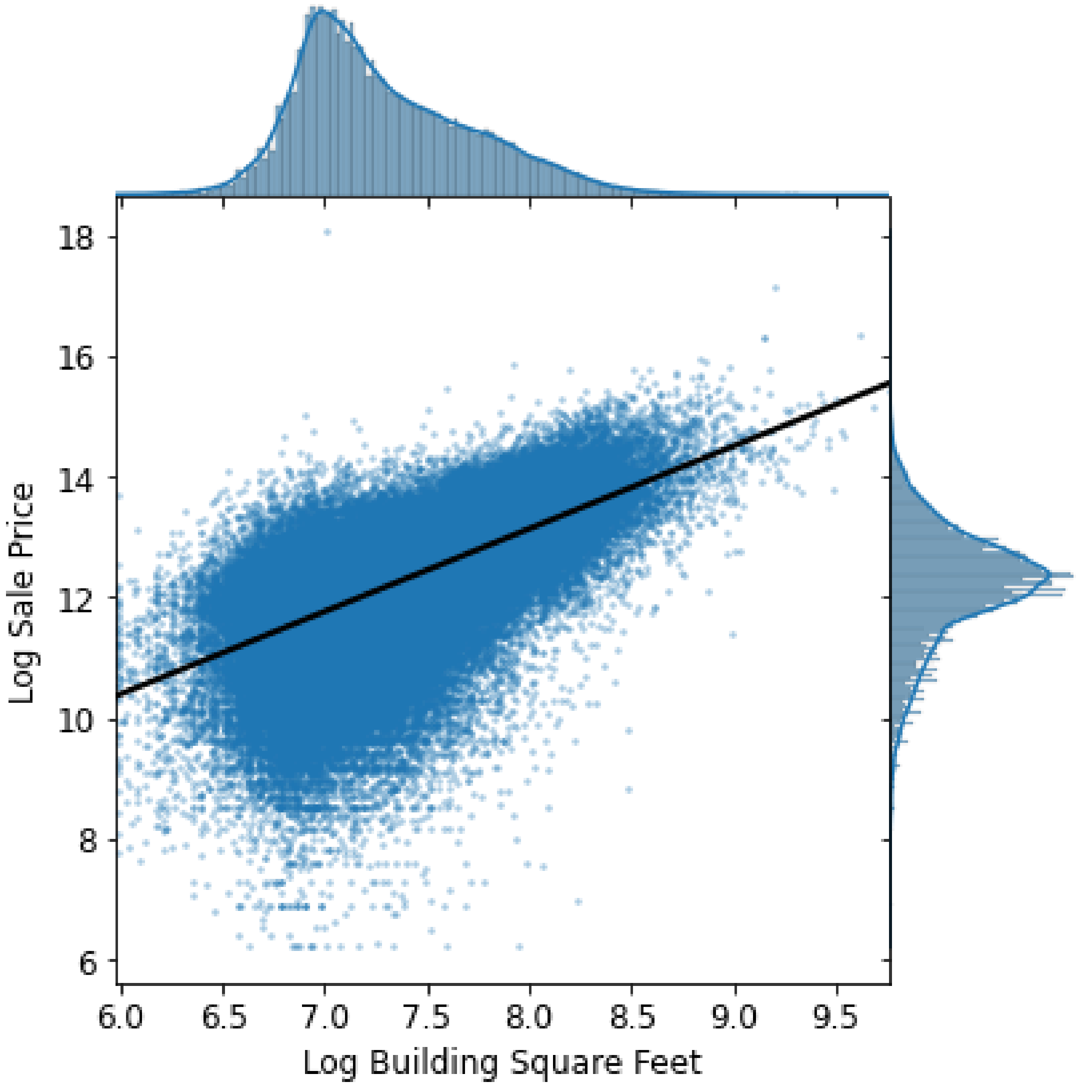
Identify one issue with the visualization above and briefly describe one way to overcome it. You may also want to try running `training_data['Sale Price'].describe()` in a different cell to see some specific summary statistics on the distribution of the target variable. Make sure to delete the cell afterwards as the autograder may not work otherwise.

There is overplotting, and we cannot really see the distribution/box plots of the graphs very well. They are both narrow. One way to overcome it is to set the range to something more managable or remove outliers. Additionally, like shown in the next problem, we can take the log transformation for better visualization.

1.2.2 Part 3

As shown below, we created a joint plot with `Log Building Square Feet` on the x-axis, and `Log Sale Price` on the y-axis. In addition, we fit a simple linear regression line through the bivariate scatter plot in the middle.

Based on the following plot, does there exist a correlation between `Log Sale Price` and `Log Building Square Feet`? Would `Log Building Square Feet` make a good candidate as one of the features for our model?



Yes, there is some correlation. It generally follows a linear trend with larger variance at the bottom left. Higher sale prices lead to smaller building square feet. But the smaller the sale prices get, the more variance and outliers there are. The general trendm, despite the outliers would make `Log Building Square Feet` a good feature for our model.

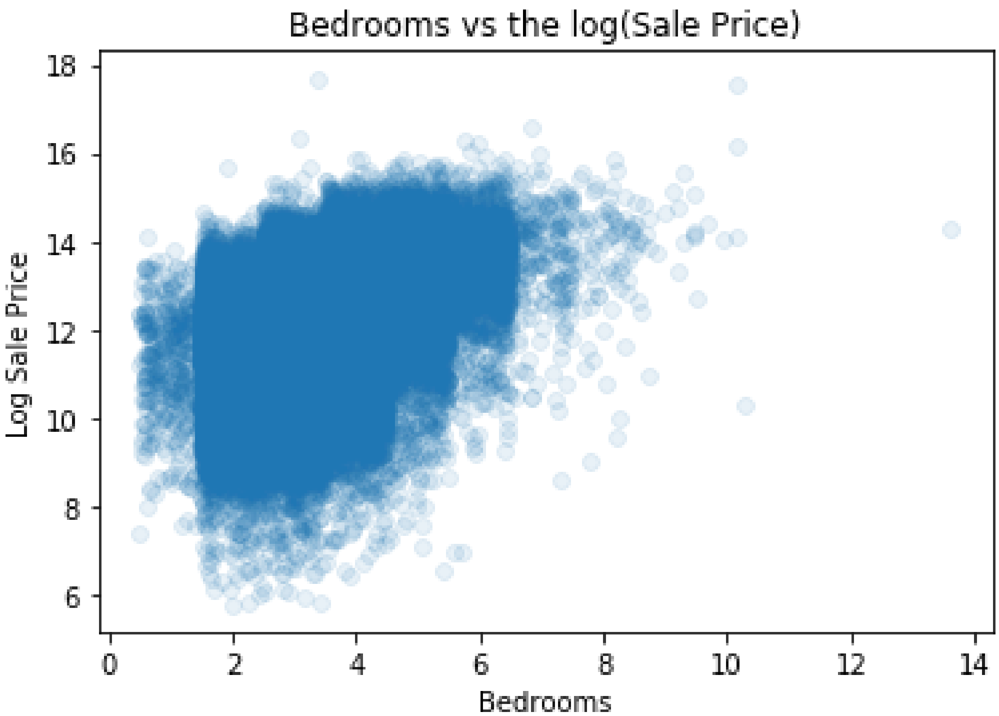
1.2.3 Part 3

Create a visualization that clearly and succintly shows if there exists an association between `Bedrooms` and `Log Sale Price`. A good visualization should satisfy the following requirements: - It should avoid overplotting. - It should have clearly labeled axes and succinct title. - It should convey the strength of the correlation between the sale price and the number of rooms.

Hint: A direct scatter plot of the sale price against the number of rooms for all of the households in our training data might risk overplotting.

[35]:
```
sns.regplot(data=training_data, x='Bedrooms', y='Log Sale Price',␣
 ↪fit_reg=False, x_jitter=0.5, y_jitter=0.5, scatter_kws={'alpha': 0.1})
plt.title("Bedrooms vs the log(Sale Price)")
```

[35]:
```
Text(0.5,1,'Bedrooms vs the log(Sale Price)')
```



1.2.4 Part 3

It looks a lot better now than before, right? Based on the plot above, what can be said about the relationship between the houses' `Log Sale Price` and their neighborhoods?

The median of log sale price is nearly the same across the top 20 neighborhoods. There is really no pattern in the spread of the prices though. Seems like an arbitrary relationship of the quartile ranges and outliers, and I hope that the next questions can give context to the relation of the neighborhood code to a certain area.

1.2.5 Part 2

Without running any calculation or code, complete the following statement by filling in the blank with one of the comparators below:

$$\geq$$
$$\leq$$
$$=$$

Suppose we quantify the loss on our linear models using MSE (Mean Squared Error). Consider the training loss of the 1st model and the training loss of the 2nd model. We are guaranteed that:

Training Loss of the 1st Model________Training Loss of the 2nd Model

$\geq$

1.2.6 Part 6

Let's compare the actual parameters (θ_0 and θ_1) from both of our models. As a quick reminder,

for the 1st model,

$$\text{Log Sale Price} = \theta_0 + \theta_1 \cdot (\text{Bedrooms})$$

for the 2nd model,

$$\text{Log Sale Price} = \theta_0 + \theta_1 \cdot (\text{Bedrooms}) + \theta_2 \cdot (\text{Log Building Square Feet})$$

Run the following cell and compare the values of θ_1 from both models. Why does θ_1 change from positive to negative when we introduce an additional feature in our 2nd model?

[53]:
```
# Parameters from 1st model
theta0_m1 = linear_model_m1.intercept_
theta1_m1 = linear_model_m1.coef_[0]

# Parameters from 2nd model
theta0_m2 = linear_model_m2.intercept_
theta1_m2, theta2_m2 = linear_model_m2.coef_[0]

print("1st Model\n 0: {}\n 1: {}".format(theta0_m1, theta1_m1))
print("2nd Model\n 0: {}\n 1: {}\n 2: {}".format(theta0_m2, theta1_m2,␣
 ↪theta2_m2))
```

```
1st Model
 0: [10.5717254]
 1: [0.49691975]
2nd Model
 0: [1.93396332]
 1: -0.030647249803554506
 2: 1.4170991378689641
```

θ_1 might be scaled down from a positive to a lower negative number because it's feature might be less relevant. Being close to 0 nullifies a feature in this linear regression model. Also it might be scaled down because we introduced a new feature in model 2.

1.2.7 Part 7

Another way of understanding the performance (and appropriateness) of a model is through a residual plot.

In the cell below, use `plt.scatter` to plot the predicted `Log Sale Price` from **only the 2nd model** against the original `Log Sale Price` for the test data. You should also ensure that the dot size and opacity in the scatter plot are set appropriately to reduce the impact of overplotting.

[54]:
```
plt.scatter(x=y_test_m2, y=y_predicted_m2, alpha=0.1, s=100)
plt.title("Log Sale Price: Test vs. Predicted")
plt.xlabel("Test dataset")
plt.ylabel("Model predicted")
```

[54]:
```
Text(0, 0.5, 'Model predicted')
```